USING ARTIFICIAL INTELLIGENCE TO IDENTIFY KEY PERFORMANCE DRIVERS IN MAJOR LEAGUE BASEBALL

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Abstract:

Identifying the metrics that drive success in Major League Baseball remains a key challenge in sports analytics. This study applies artificial intelligence (AI) to uncover key performance indicators and to support data-driven decision-making using MLB Statcast data in professional baseball. Using five years of Statcast data, we applied AI-driven methods, including K-means clustering, multiple regression, and Random Forest models, to identify performance patterns and predict player success in a professional sports context. K-means clustering identified distinct groups of hitters, where high-value players combined higher on-base performance with more selective swing behavior, while lower-value players showed weaker discipline and production.

Regression results explained 55.5% of the variance in players' Wins Above Replacement (WAR) ($R^2 = 0.555$, $p < .001$), indicating predictive power, though a significant portion of performance remained unexplained. Slugging percentage emerged as the strongest predictor, followed by contact on pitches outside the strike zone, while poor swing decisions negatively impacted performance. However, because a significant portion of performance remains unexplained, overreliance on AI may overlook contextual factors such as player health and situational performance, emphasizing the need to use these models as decision-support tools rather than replacements for human judgment. These findings demonstrate how AI can help MLB teams identify key performance drivers and improve strategic, data-driven decision-making, yet human oversight is still needed to account for unencoded contextual and situational factors not captured in the model.

Background Information:

Statcast provides detailed, high-frequency data on player behavior, including swing decisions, contact rates, and power metrics. These variables allow for deeper insights into key aspects of hitting performance, such as plate discipline and power production.

Artificial intelligence (AI) methods, such as clustering and machine learning, have emerged as powerful tools for identifying patterns in large datasets (Kuhn, 2008; Liaw & Wiener, 2002). These approaches can uncover complex relationships between performance variables that are not easily detected using traditional statistical methods (Kuhn, 2008).

This study applies AI-driven techniques to determine which offensive metrics most strongly influence player value, measured by Wins Above Replacement (WAR), and evaluates how these tools can support data-driven decision-making in professional baseball.

Methods/Data:

- **Data Source:** MLB Statcast data (2021–2023)
  - 122 players, 176 performance variables
  - Focus on plate discipline, contact ability, and power metrics
- **Analytical Tools:** RStudio (data cleaning, modeling, visualization)
- **K-Means Clustering**
  - Variables: OBP, WAR, Z-Swing%
  - Purpose: Identify distinct hitter profiles
- **Random Forest Model**
  - Used to determine variable importance
  - Dataset split into training/testing sets
  - Identified most influential predictors of offensive success
- **Multiple Linear Regression:** Evaluated using R^2 and significance testing
  - **Dependent variable:** WAR
  - **Independent variables:** SLG (Slugging %), O-Swing%, O-Contact%, Z-Swing%, Z-Contact%

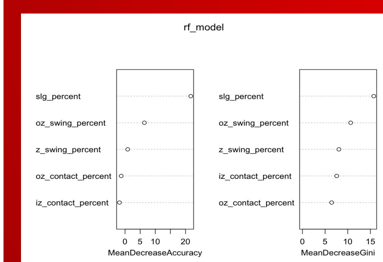

Figure 1: 2021 Random Forest model

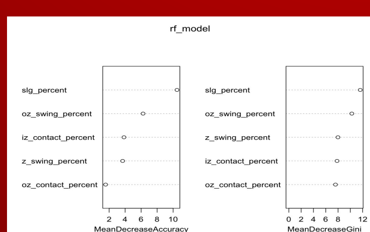

Figure 2: 2022 Random Forest model

Credit Authorship Contribution Statement:

Buchanan, K.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data Curation, Writing, RStudio coding; **Kirkpatrick, S.**: Methodology, Formal analysis, Investigation, Review & Editing; **Stoeckel, F.**: Conceptualization, Methodology, Software, Visualization, Graphics, Review & Editing; **Marsellos, A.E.:** Supervision, Project administration, Guidance, Editing; **Yetgin, E.**: Supervision, Project administration, Guidance, Editing.

Results:

Clustering Findings

- Four distinct hitter profiles identified
- High-value players: Higher OBP, better plate discipline (selective swinging)
- Lower-value players: Poor discipline, lower offensive production

Regression Results

- Model significance: $F(5,455) = 113.68$, $p < .001$
- Explained variance: $R^2 = 0.555$

Key Predictors

- Slugging % (SLG) → strongest positive predictor
- O-Contact% → positive effect
- O-Swing% → negative effect

Simplified Model

- Similar performance ($R^2 = 0.553$)
- Confirms: Power hitting and plate discipline are primary drivers of success

Overall Insight

Players who hit for power, avoid chasing bad pitches, still make contact when they do
→ produce the highest value (WAR)

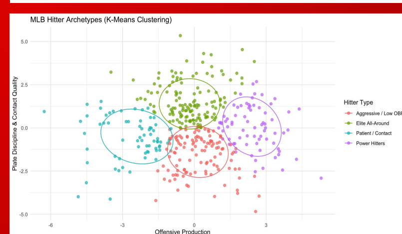

Figure 3: 2021-2023 k-means clustering plate discipline & contact quality by offensive production clustered by hitter type

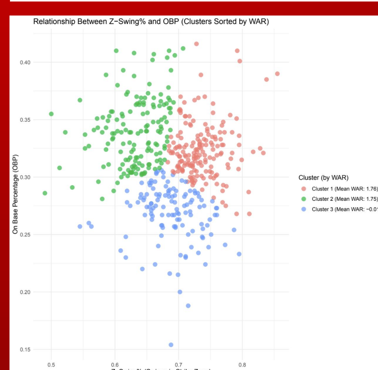

Figure 4: 2021-2023 k-means clustering relationship between Z-Swing% and OBP clustered by WAR

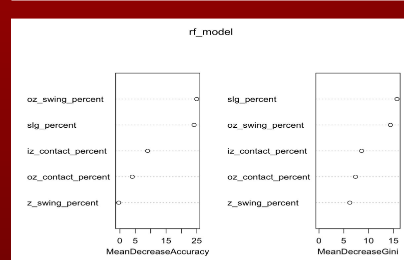

Figure 5: 2023 Random forest model

Discussion:

Artificial intelligence is playing a growing role in identifying key MLB performance drivers. Across models, **slugging percentage (SLG)** consistently emerged as the strongest predictor of offensive success, reinforcing the value of power hitting.

Plate discipline was also critical: lower **O-Swing%** and higher **O-Contact%** significantly improved performance. Successful hitters combine power with selectivity and adaptability at the plate.

K-means clustering revealed that high-value players blend strong on-base skills with disciplined swing behavior. These findings support modern analytics focused on efficiency and smart decision-making.

The models explained only **55% of WAR variance**, leaving nearly half of player performance unexplained. Factors like health, defense, and situational play likely account for the rest. AI excels at pattern detection but cannot replace human judgment.

Key Takeaway: AI should support, not replace, scouting and coaching expertise.

Conclusion:

This study demonstrates that AI effectively uncovers key performance drivers in MLB using Statcast data. **Slugging percentage**, plate discipline, and outside-zone contact ability were the strongest predictors of offensive success.

However, with **45% of performance variability unexplained**, statistical models have clear limits. AI analytics should complement traditional evaluation methods.

Future Directions:

- Include defensive metrics, injury history, and situational data
- Explore advanced modeling techniques
- Integrate AI with human scouting and coaching

Combining data-driven insights with human expertise will help teams gain a competitive edge.

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References:

Brennan, B. (2022). *Statcast data: Hitters 2022* [Dataset]. Kaggle. https://www.kaggle.com/datasets/brendanstyle/statcast-data-hitters-2022/data

Kuhn, M. (2008). Building Predictive Models in R Using the caret Package. Journal of Statistical Software, 28(5), 1–26. https://doi.org/10.18637/jss.v028.i05

Liaw A, Wiener M (2002). "Classification and Regression by randomForest." _R News_, *2*(3), 18-22. <https://CRAN.R-project.org/doc/Rnews/>.

Maechler, M., Rousseeuw, P., Struyf, A., Hubert, M., Hornik, K.(2026). cluster: Cluster Analysis Basics and Extensions. R package version 2.1.8.2. Kassambara A, Mundt F (2020). _factoextra: Extract and Visualize the Results of Multivariate Data Analyses_. doi:10.32614/CRAN.package.factoextra <https://doi.org/10.32614/CRAN.package.factoextra>, R package version 1.0.7, <https://CRAN.R-project.org/package=factoextra>.

Mossey, M. (n.d.). *Major League Baseball hitting data* [Dataset]. Kaggle. https://www.kaggle.com/datasets/m000sey/major-league-baseball-hitting-data

Schroeder, E. (n.d.). *Statcast data from 2021 to 2024* [Dataset]. Kaggle. https://www.kaggle.com/datasets/ericschroeder1/statcast-data-from-2021-to-2024

Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R, Grolemund G, Hayes A, Henry L, Hester J, Kuhn M, Pedersen TL, Miller E, Bache SM, Müller K, Ooms J, Robinson D, Seidel DP, Spinu V, Takahashi K, Vaughan D, Wilke C, Woo K, Yutani H (2019). "Welcome to the tidyverse." _Journal of Open Source Software_, *4*(43), 1686. doi:10.21105/joss.01686 <https://doi.org/10.21105/joss.01686>.

Wickham H, François R, Henry L, Müller K, Vaughan D (2023). _dplyr: A Grammar of Data Manipulation_. doi:10.32614/CRAN.package.dplyr <https://doi.org/10.32614/CRAN.package.dplyr>, R package version 1.1.4, <https://CRAN.R-project.org/package=dplyr>.